Farnaz Rezvani, MSc

Project Coordinator, Pitt HexAI Research Laboratory
GitHub Lead, Pitt HexAI Research Laboratory
Research Assistant, Pitt HexAI Research Laboratory
Research Assistant, Medical Innovation Research in AI Laboratory
IEEE Member, AMIA Member
London, UK

Page: Google Scholar
Page: LinkedIn Profile
Page: Pitt HexAI Profile
Page: Medical Innovation Research
Page: ORCiD
E-mail: f.rezvani@ieee.org
Phone: +447983300426

Research Interests

- **Technical:** Artificial Intelligence (AI), Machine Learning (ML), Deep Learning, Trustworthy and Explainable AI, Computer Vision, AI for Medical Imaging, Vision Foundation Models, Vision-Language Models, Federated Learning, Survival Analysis
- **Application:** Healthcare AI, Clinical Risk Prediction, Healthcare Informatics, Musculoskeletal and Orthopedic Artificial Intelligence, Medical Decision Support, Postoperative Outcome Prediction, Length-of-Stay Prediction

Summary

- 15+ years of professional experience in software engineering, database systems, data analytics, artificial intelligence, and machine learning across industry and academic environments.
- Research focused on trustworthy, explainable, and uncertainty-aware AI for healthcare, with expertise in clinical risk prediction, medical imaging AI, computer vision, federated learning, and vision foundation models.
- Developed AI systems for clinical decision support, postoperative complication prediction, length-of-stay prediction, and musculoskeletal disease analysis using machine learning, deep learning, federated learning, and survival analysis.
- First author of peer-reviewed journal and conference publications in trustworthy healthcare AI, with research contributions spanning predictive modeling, medical imaging, explainable AI, and federated learning.
- Project Coordinator and GitHub Lead at the University of Pittsburgh Pitt HexAI Research Laboratory, coordinating collaborative AI research projects, mentoring junior researchers, and supporting reproducible AI research workflows.

Education

2023–2024	**MSc**, Data Science and Analytics, Brunel University London, UK
2007–2009	**BSc**, Computer Science, Industrial Management Institute, Iran

Appointments

Jun. 2026–Present	**Project Coordinator**, Pitt HexAI Research Laboratory, University of Pittsburgh, USA. (Volunteer)
Jul. 2025–Present	**GitHub Lead**, Pitt HexAI Research Laboratory, University of Pittsburgh, USA. (Volunteer)
Oct. 2024–Present	**Research Assistant**, Pitt HexAI Research Laboratory, University of Pittsburgh, USA. (Volunteer)
May. 2026–Present	**Research Assistant**, Medical Innovation Research in AI Laboratory, Turkey. (Volunteer)
Apr. 2018–Jun. 2023	**Data Analyst**, Amiran Radman Co., Iran
Sep. 2007–Oct. 2014	**Database Administrator**, Gardan Part Sazan Co., Iran

Technical Skills

Healthcare AI	Clinical risk prediction, healthcare informatics, orthopedic outcomes modeling, musculoskeletal AI, medical decision support
Trustworthy AI	Explainable AI (SHAP, LIME, Grad-CAM, Integrated Gradients), uncertainty quantification, model calibration, confidence estimation
Medical Imaging AI	Radiograph analysis, medical image preprocessing, YOLO-based ROI detection, computer vision, vision foundation models (CLIP, DINOv2), vision-language models
Federated Learning	FedAvg, FedProx, federated neural networks, federated XGBoost, non-IID learning
Survival Analysis	Time-to-event prediction, Cox regression, DeepHit, survival analysis, length-of-stay prediction
Machine Learning	Scikit-learn, XGBoost, Random Forest, Logistic Regression, Deep Learning, CNNs

Programming Python, R, SQL, PySpark

Tools GitHub, VS Code, Google Colab, Hugging Face, Flower, Tableau, Power BI

Peer-Reviewed Publications

[1] **Rezvani F**, Kann MR, Gao F, Gupta P, Myers N, Kashefi A, Visweswaran S, Wang Y, Hogan MV, Plate JF, Tafti AP. Trustworthy AI models to predict prolonged hospital length-of-stay in orthopedic surgery. npj Health Systems. 2026 Jul 1;3(1):56. [Link]

[2] **Rezvani F**, Towsen K, Menezes Z, Einhorn A, Davis J, Gupta P, Plate JF, Fox C, Myers N, Tafti AP. Trustworthy and Uncertainty-Aware AI for Predicting Respiratory Complications Following Total Hip and Knee Arthroplasty. In AMIA Annual Symposium Proceedings 2025 May 22 (Vol. 2024, p. 1089). [Link]

[3] Gao F, **Rezvani F**, Han Y, Kann M, Littlefield N, Maradit Kremers H, Yates AJ, Plate JF, Tafti AP. Uncertainty-quantified and explainable age- and sex-aware contrastive learning for knee osteoarthritis classification. In: Proceedings of the International Symposium on Biomedical Imaging (ISBI); 2026. [Link]

[4] Han Y, Gao F, **Rezvani F**, Littlefield N, Kann MR, Shah P, Myers N, Plakseychuk E, Heidinger EM, Ewalefo S, Hogan M, Plate JF, Tafti AP. Label-efficient data engineering for reliable orthopedic radiograph registry construction under distribution shift. In: Data Engineering in Medical Imaging (DEMI), MICCAI Workshop; 2026. [Accepted]

[5] Bogden E, **Rezvani F**, Bruno E, Poorhasani P, Farahani S, Agharazidermani M, Mathew J, Urish KL, Plate JF, Ghazaleh H, Tafti AP. Explainable federated learning to predict cardiac complications following total hip and knee arthroplasty. In: AMIA Annual Symposium; 2026. [Accepted]

[6] Menezes Z, **Rezvani F**, Einhorn A, Fox C, Towsen K, Mathew J, Kann MR, Myers N, Gupta P, Cong T, Urish KL, Yates AJ, Plate JF, Tafti AP. Explainable machine learning with uncertainty quantification for cardiac risk prediction following total knee and hip arthroplasty. In: AMIA Annual Symposium; 2026. [Accepted]

Abstract Publications

[1] **Rezvani F**, Jain V, Gao F, Littlefield N, Kann M, Plate JF, Tafti AP. Prompt-constrained vision-language models for zero-shot orthopedic pathology recognition. In: Proceedings of the Interna- tional Symposium on Biomedical Imaging (ISBI); 2026.

[2] **Rezvani F**, Kann M, Gao F, Myers N, Plaseychuk E, Hogan MV, Tafti AP, Plate JF. Preoperative trustworthy and explainable AI for early length-of-stay prediction in foot and ankle surgery. Presented at: American Orthopaedic Foot & Ankle Society (AOFAS) Annual Meeting; 2026.

[3] **Rezvani F**, Kann MR, Shah P, Myers N, Visweswaran S, Tafti AP, Urish KL, Plate JF. Are state-of-the-art machine learning and/or large language models (LLMs) able to predict infection risks after total knee arthroplasty using only pre-operative variables? In: Joint EBJIS–MSIS 2026 Conference; 2026.

[4] Gao F, **Rezvani F**, Kann M, Littlefield N, Pitt W, Irrgang JJ, Yates AJ, Plate JF, Tafti AP. An explainable age- and sex-aware contrastive artificial intelligence framework for knee osteoarthritis classification. In: Proceedings of the Interna- tional Symposium on Biomedical Imaging (ISBI); 2026.

[5] Han Y, Gao F, **Rezvani F**, Littlefield N, Kann MR, Shah P, Myers N, Plaseychuk E, Heidinger EM, Ewaleko S, Hogan MC, Plate JF, Tafti AP. A maintainable and uncertainty-aware AI-powered pipeline for knee radiograph registry labeling under domain shifts. In: AMIA Annual Symposium Proceedings. 2026. [Accepted]

Selected Projects

- Trustworthy AI for post-arthroplasty respiratory risk prediction [GitHub Link]
- Cardiac risk prediction after THA/TKA using explainable and uncertainty-aware machine learning models [GitHub Link]
- Cervical cancer risk prediction using supervised machine learning [GitHub Link]
- HexAI Ortho: An AI-powered documentation tool set in orthopaedic setting [GitHub Link]
- HexAI Tibia: An AI-driven tool set to measure tibia slope in radiographs [GitHub Link]
- Uncertainty-Quantification and ML explainability for cervical cancer classification
- Airbnb price forecasting using machine learning and PySpark [GitHub Link]
- Global superstore sales visualization using interactive dashboards in Tableau/Power BI [GitHub Link]
- Survival trend analysis on SEER cancer dataset using distributed MapReduce framework in Python [GitHub Link]

Certifications

- **Stanford University (2025):** Fundamentals of AI/ML in Precision Medicine
- **Udemy (2025):** Modern Computer Vision
- **Udemy (2025):** Federated Learning
- **FreeCodeCamp (2025):** Data Analysis with Python
- **TimeXtender (2025):** Data Quality Academy
- **WPHIMA (2024):** Explainable AI in Healthcare
- **LinkedIn Learning (2023):** SQL and Python for Data Analysis
- **LinkedIn Learning (2023):** Python for Data Analysis

Professional Activities and Operational Services

- ACM TRUST 2027

  Role: Web Master

- University of Pittsburgh AI Summer School [Link]

  Role: Keynote Speaker
  Role: Web Master

- IEEE Access (2025 - Present)

  Role: Reviewer/Referee

- AMIA (American Medical Informatics Association) (2025 - Present)

  Role: Reviewer/Referee

- Responsible, Explainable, and Ethical-by-Design Artificial Intelligence in Clinical Applications (REED-AI) 2026

  Role: Reviewer/Referee
  Role: Web Master

Mentorship and Supervision

- **Mentor**: Vivaan Jain, Research Intern from Delhi Technological University (DTU), New Delhi, India; supervised his work on vision-language models for orthopedic imaging (Oct 2025).

Memberships in Professional and Scientific Societies

Organization	**Year**
Institute of Electrical and Electronics Engineers (IEEE)	2024–Present
American Medical Informatics Association (AMIA)	2026–Present